MIL-B-81744A

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MILITARY SPECIFICATION

BARRIER COATING SOLUTION, LUBRICANT MIGRATION DETERRING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for a lubricant-migration-deterring barrier coating solution consisting of a fluoropolymer identified as poly (1H,1H-pentadecafluorooctyl methacrylate) dissolved in xylene hexafluoride.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-763	Steel Bars, Shapes, and Forgings-Corrosion Resisting

Military

MIL-S-81087	Silicone Fluid, Chlorinated Phenyl Methyl Polysiloxane
MIL-B-81744A/1	Barrier Coating Solvent

STANDARDS

Federal

Fed. Test Method Std. No. 141	Paint, Varnish, Lacquer; and Related Materials: Methods of Inspection, Sampling and Testing

FSC 6850

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STANDARDS (Cont'd)

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1334	Process for Barrier Coating of Anti-Friction Bearings

(Copies of specifications and standards required by suppliers in connection with procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TEST AND MATERIALS PUBLICATIONS
ASTM Standards Parts 17 and 18

(Copies of ASTM Publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Materials - The barrier coating solution shall be composed of a 2 per cent (by weight) solution of poly 1H,1H-pentadecafluorooctyl methacrylate in xylene hexafluoride (conforming to MIL-B-81744A/1 to which has been added ultraviolet indicator Tenapol SSG (see 6.3) at a concentration of 0.2 per cent based on weight of the fluoropolymer.

3.2 Abrasion - A circle of barrier film prepared as specified in 4.2.3 and tested as specified in 4.2.4 shall, after being subjected to one abrasive stroke, prevent a drop of chlorinated silicone lubricant conforming to MIL-S-81087 from passing across or through the film for 24 hours.

3.3 Contact angle - A drop of n-hexadecane shall form a contact angle of not less than 60 degrees when a film of the barrier coating is tested as specified in 4.2.5.

3.4 Thermal stability - A circle of barrier film when tested as specified in 4.2.6 at 347°F ±3.6°F (175 ±2°C) shall prevent the spread of a drop of the silicone from passing outside the barrier film for 24 hours.

3.5 Contamination in solution - When tested as specified in 4.2.7 a distilled water drop placed in the center of the circle of barrier film after the reagent grade acetone has evaporated shall spread and not form a bead.

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3.6 Indicator - When tested as specified in 4.2.8 the test surface shall fluoresce under black light.

3.7 Nonvolatile content - When the barrier coating solution is tested in accordance with 4.2.9, the nonvolatile content shall be 2.0 ±0.1 grams.

3.8 Workmanship - The solution, when examined visually shall be clear, homogeneous, and free from suspended matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified the supplier may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection -

4.2.1 Sampling - The quality conformance sample shall consist of a sample for tests (4.2.1.3) and a sample for examination of filled containers (4.2.1.4).

4.2.1.1 Lot - A lot of polymer (employed in the preparation of the solution) or solvent (employed in the preparation of solution) is defined as the yield from a single series of chemical operations leading to production of the ingredients used to make the barrier coating solution.

4.2.1.2 Batch - A batch of the barrier coating solution is defined as the yield from a single series of operations, utilizing the same combination of solvent and polymer. A change in either solvent or polymer shall constitute a new batch of solution.

4.2.1.3 Samples for test - Before being placed in unit containers, two 10 ml samples from each batch of finished solution shall be taken. Each sample shall be identified by batch number, source of supply and date of manufacture. The batch shall be unacceptable if a sample fails to meet any of the test requirements specified in Section 3. In addition, after filling and capping, each glass unit container (the same container in which the material is to be delivered) of solution shall be sampled, identified as above and tested for conformance with the test requirement of 4.2.7. The material contained in the unit container shall be unacceptable if the sample fails to meet this test requirement.

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4.2.1.4 Sample for examination of filled containers - All filled shipping containers fully prepared for delivery from each batch shall be subjected to examination.

4.2.2 Inspection - Quality conformance inspection shall consist of an examination for acceptability of quality control methods used by the manufacturer, an examination of samples of filled containers (4.2.1.4) for conformance to packaging packing, and marking requirements, and examination and tests of samples (4.2.1.3) for all the requirements specified in Section 3.

4.2.3 Method for preparing film - Unless otherwise specified, the following procedure shall be used to prepare films to be tested for conformance with the requirements of this specification. A finish ground platen of stainless steel conforming to Class 302 or 304 of QQ-S-763 shall be abraded under running tap water with No. 600 waterproof silicon carbide paper and "Tide" detergent (or equivalent) until a random pattern has replaced the oriented grinding marks. The platen shall then be flushed with distilled water to remove all traces of abrasive and detergent. (At this point it is important to verify that distilled water drains smoothly from the surface.) There shall be no area of test surface on which the water will draw up into droplets or pools, as if the surface were greasy. After the distilled water flush the platen, while still wet with water shall be held vertically and flushed with reagent grade acetone to remove the remaining water and cause rapid drying of the surface. (Proof of the purity of the reagent grade acetone can be established after the acetone drying step by placing a sample platen in a horizontal plane and contacting it in the center with a large drop of distilled water. If the drop maintains a hemispherical or dome shape the platen has an organic hydrophobic contaminating film present. Such films probably result from impure acetone and will adversely affect the validity of test methods described below). Test rings of the barrier film shall be 0.5 to 0.75 inches outside diameter and the barrier band shall be about 0.1 inch wide. The film forming solution shall be applied with a small lintless tissue swab held with tweezers or with a small piece of lintless tissue packed into the tip of a medicine dropper from the back side. The application should be made by only two circular passes with a damp (not dripping wet) swab. Care must be taken to avoid building a thick film, because thicker films are less adherent and are more easily dislodged. Satisfactory films should show a characteristic violet fluorescence when the film is illuminated by an ultraviolet light source described in 4.2.8. Viewing under ultraviolet light will permit detection of holidays in the film which would prevent the barrier film from containing a lubricant. Films shall be allowed to air dry at least 30 minutes before they are subjected to test. Three rings shall be used for each of the tests specified herein. If at least one of the three replicates pass a test, the sample shall be considered satisfactory for this test.

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4.2.4 Abrasion - Using circles of barrier film as prepared in 4.2.3, the tip of the index finger shall be smoothly wrapped in fresh lens tissue, and placed firmly with a force of 2 to 3 pounds against the uncoated surface in the center of the ring and drawn, under the same pressure, against the barrier band onto the clean metal outside. The lens tissue from this ring shall not be used for another stroke or across another test ring, and only one stroke shall be made across each barrier film. After all test circles have been subjected to the abrasion stroke a small drop of MIL-S-81087 silicone shall be placed in the center of the test circles and the platens stood vertically with abraded sections toward the bottom. The oil should flow downward to the margin of the barrier ring and be retained at this point. If, during a period of 24 hours, oil passes across or lifts the barrier, the barrier material shall be considered to have failed this test.

4.2.5 Contact angle - The contact angle formed by a drop of n-hexadecane on the barrier film formed on a clean microscope slide by dipping it into the solution shall be read with a contact angle goniometer or by means of an eyepiece goniometer in a microscope fitted with low power (2X or 3X) objective. The drop of n-hexadecane shall be formed from the end of a clean 0.025 inch diameter nichrome wire. If the above apparatus is not available the angle may be observed with sufficient accuracy either with an optical comparator or through the use of a Polaroid micrograph showing the cross section outline and the plane on which it rests and using a protractor to measure the angle through the liquid.

4.2.6 Thermal stability - Using circles of barrier films as prepared in 4.2.3 a drop of silicone conforming to MIL-S-81087 and formed from the end of a clean 0.025 inch diameter nichrome wire shall be placed in the center of separate circles. The samples shall be covered with petri dishes and stored horizontally in a convection oven controlled at 347 $\pm$3.6°F (175 $\pm$2°C) for 24 hours. Observations shall be made to see if any fluid is outside the barrier film circle.

4.2.7 Contamination in solution - Each filled and capped unit container of solution (see 4.2.1.3) shall be thoroughly shaken with cap in place. The cap shall then be removed and a few drops poured out onto a clean lintless swab. The cap shall then be replaced, taking care not to introduce contamination. Using the swab to apply circles of barrier film (as in 4.2.3) to acid cleaned glass slides, a pool of reagent acetone shall be built up inside the circles using a medicine dropper for initial insertion and for maintaining the pool against evaporation losses in 5 minutes. Then the acetone shall be permitted to evaporate to dryness, and a drop of distilled water added to center of the circle. Observation shall be made to determine if the water drop stands as a bead or whether it spreads.

4.2.8 Ultraviolet fluorescent indicator - Using a clean microscope slide as prepared in 4.2.5 place under an ultraviolet lamp emitting

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either 2537 Angstrom or 3660 Angstrom line in an otherwise unilluminated darkroom, in order to observe violet fluorescence.

4.2.9 Nonvolatiles - The nonvolatiles of the barrier coating solution shall be determined in accordance with Method 4045 of Federal Test Method Standard 141.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Unless otherwise specified packaging shall be Level A or C (see 6.2).

5.1.1 Level A - Unless otherwise specified the barrier coating solution shall be packaged in wide mouthed clear glass bottles or epoxy-lined metal containers. Screwcaps shall be polyethylene or polypropylene, with a layer of aluminum foil.

5.1.2 Level C - The barrier coating shall be packaged to afford the minimum degree of protection necessary to prevent damage during shipment under environmental conditions and commercial modes of transportation.

5.2 Packing - Unless otherwise specified, packing shall be Level C (see 6.2).

5.2.1 Level C - Packages that require overpacking for acceptance by the common carrier shall be packed in exterior type shipping containers in a manner that will assure safe transportation at the lowest cost to the point of delivery and shall meet as a minimum the requirements of the rules and regulations of the mode of transportation selected.

5.3 Marking - Marking shall be in accordance with MIL-STD-129. In addition to any special marking required by the contract or order, the unit containers shall be marked with the following information:

Barrier Coating Solution, Lubricant Migration Deterring
Specification MIL-B-81744A
Contract No.
Batch No.
Lot No.
Caution: This solution may interfere with any subsequent lubrication of surfaces with which it comes in contact. Particular care should be taken to avoid contamination of work areas and cleaning solutions. See MIL-STD-1334 for additional information.

6. NOTES

6.1 Intended use - The solution covered by this specification is furnished as a 2 percent solution of a fluoropolymer in xylene hexafluoride and is intended for application to selected surfaces of precision ball bearings to form thin oil-repellent film which retains the lubricant within the bearing and reduces lubricant migration. The shelf life of bearings is extended significantly and longer service lives of bearings

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and bearing parts have been effected. The 2 percent concentration of the barrier coating solution should be maintained by addition of xylene hexafluoride solvent conforming to MIL-B-81744/1 to compensate for evaporation losses.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title and number of this specification

(b) Quantity desired

(c) Type and capacity of containers in which barrier coating solution is to be furnished

(d) Applicable levels of packaging, if other than as specified in 5.1

6.3 Tanapol SSG is manufactured by Geigy Chemical Corp., Ardsley, N.Y. 10502.

6.4 There is a paucity of information concerning the toxicity of xylene hexafluoride. In the absence of additional information xylene hexafluoride or xylene hexafluoride solutions should be treated like chlorinated solvents. All operations using the barrier coating solution should be accomplished under a hood, or in conjunction with other systems of forced removal of fumes.

Custodians:
Army - MU
Navy - AS
Air Force - 11

Review activities:
Army - AV, EL, MI
Navy - OS, SH
Air Force - 80

Preparing Activity:
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